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CMU-Africa

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Day map



Night query

Goal: localize reliably

Illumination

Day → night

strong appearance shift

Viewpoint

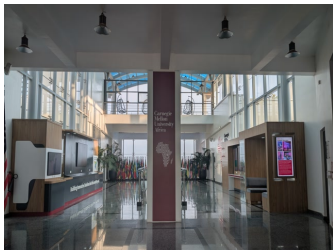
Small pose changes

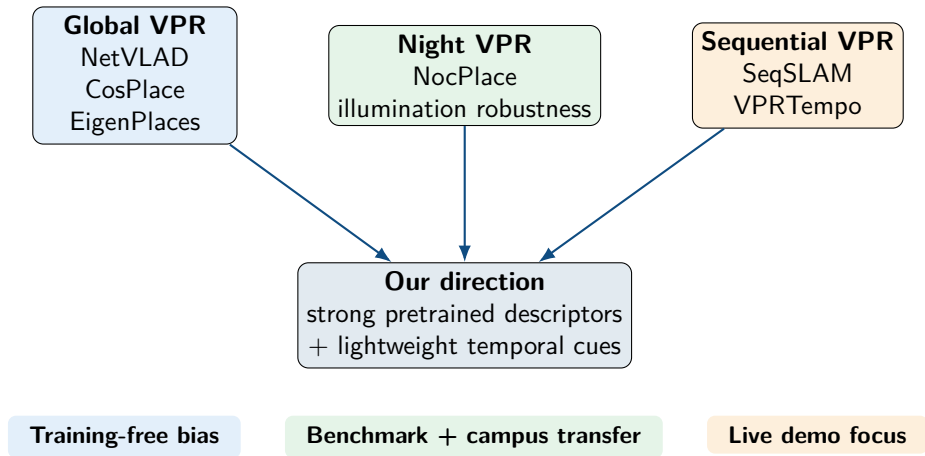
can distort ranking

Deployment

Live inference

must stay responsive





Benchmark

GardensPoint

controlled day-to-night evaluation

Metrics

AUC

Recall@1 / 5 / 10

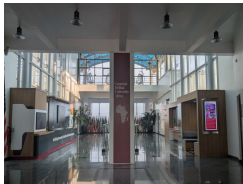
Recall@100P

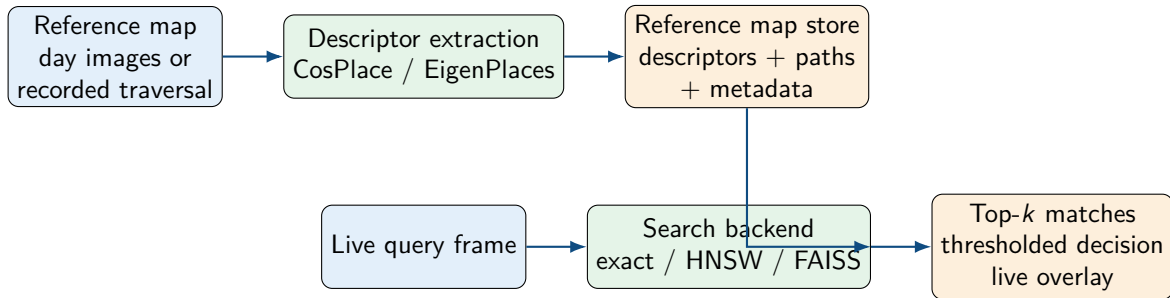
Our campus dataset

50 day reference images

64 night queries

33 matched, 31 no-match

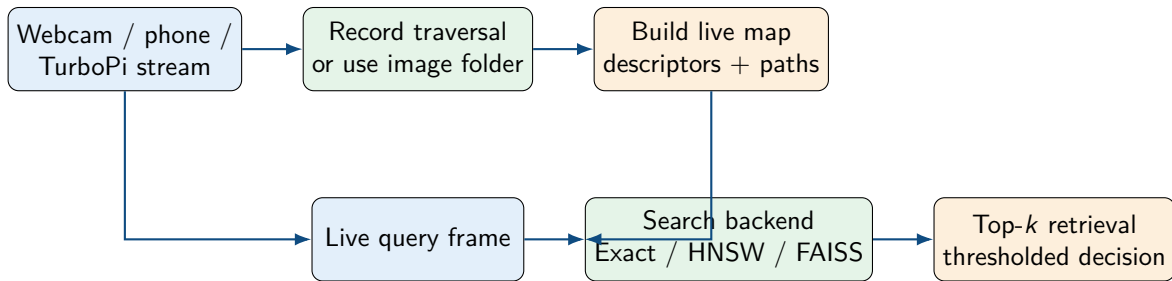




Offline: build map

Online: search + rank

Demo: webcam / phone / robot



Config-driven pipeline

Modular search layer

Ready for live demo

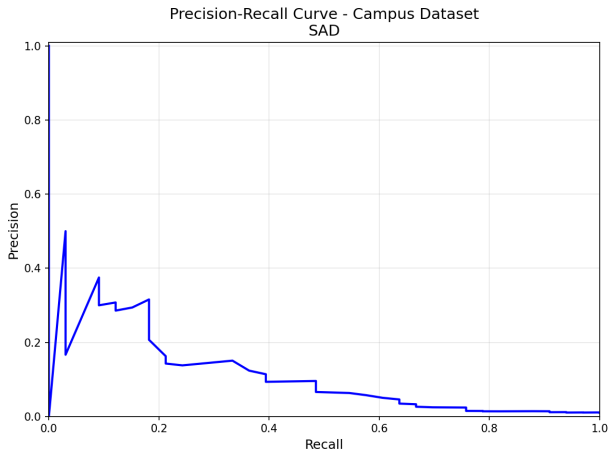
Model	AUC	R@1	R@5	R@10
CosPlace	0.592	0.350	0.840	0.925
EigenPlaces	0.676	0.370	0.840	0.940
AlexNet conv3	0.195	0.300	0.665	0.745
SAD	0.030	0.090	0.225	0.305

Best baseline: EigenPlaces

Takeaway

Top-5 is already strong

Main gap: ranking the correct place first



Campus / CosPlace

$$R@1 = 0.727$$

$$R@5 = 1.000$$

$$AUC = 0.470$$

50 day refs

64 night queries

33 matched / 31 no-match



10 place IDs

Main finding

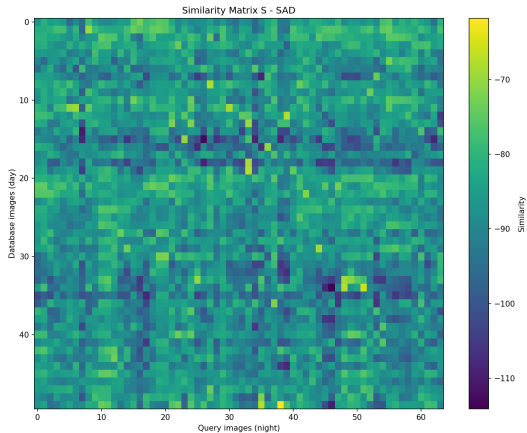
Original campus setup
was too strict for
multi-view place recognition

After place-level relabeling

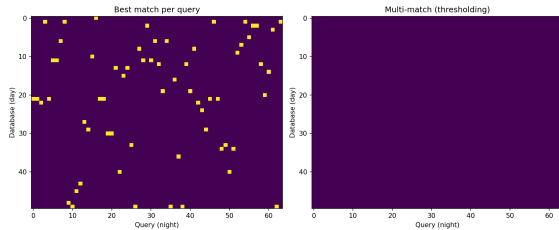
CosPlace: R@1 0.727 $\rightarrow$ 0.969

EigenPlaces: AUC 0.443 $\rightarrow$ 0.663

NetVLAD: R@1 0.394 $\rightarrow$ 0.891



Similarity structure



Correct vs wrong matches



Current practical pipeline

Build map from images or traversal video

Run live localization from webcam /
stream

Swap search backends through config

Applied CV emphasis

Robot-facing deployment

Demo-ready workflow

1

Strong benchmark
baseline

2

Campus transfer
improved after
place-level relabeling

3

End-to-end live system
is in place

Key takeaway: strict labels under-credited multi-view matches

Robust live demo

Search at larger map scale

Presentation

Apr 26, 2026

2:00 PM, F203

8 min talk + 4 min Q&A

Random presentation order

Submission

5-min video presentation

Due Apr 25, 11:59 PM

Final report + reproducible GitHub +
demo link

Due Apr 30, 11:59 PM CAT

Thank you

Demo / Questions